TASK 19: TOP 10 PRODUCTS BY SALES

Data Analytics Internship | Veda Technologies

Project Overview

This project uses the Sample Superstore dataset to identify the ten products with the highest total sales. Python, pandas, and Matplotlib are used to aggregate sales across order records, rank products, and visualize the results.

Objective

Identify the top 10 products by total sales, practice sorting and ranking, create a visualization, and understand how ties can be handled consistently.

Tools and Technologies

Python | pandas | Matplotlib | Jupyter Notebook | Sample Superstore Excel workbook

Methodology

Step	Description
1	Load the Orders worksheet from the Sample Superstore workbook.
2	Inspect the dataset and confirm the product-name and sales columns.
3	Group records by product and sum Sales to calculate total sales per product.
4	Sort products from highest to lowest total sales.
5	Select the top 10 products and assign ranks.
6	Create a horizontal bar chart to compare the selected products.
7	Export the ranked results to a CSV file.

Expected Outputs

• A ranked table of up to 10 products and their total sales.
• A horizontal bar chart comparing the selected products.
• A CSV file named *top_10_products_by_sales.csv*.

Ranking and Tie Handling

Sales are summed across all order rows for each product before ranking. To show exactly ten products, use product name as a consistent tie-breaker when sales totals are equal. If the requirement is to include all products tied at tenth place, include every product with the same cutoff sales value.

Key Learning Outcomes

• Aggregate repeated product records with pandas *groupby()* and *sum()*.
• Sort and rank products using a numerical measure.
• Present comparisons with a Matplotlib bar chart.
• Understand that sales and profit are different business measures.

Conclusion

This analysis demonstrates how Python can identify high-sales products and communicate their relative performance clearly. The ranking can support further business investigation, while profitability should be assessed separately using profit and cost measures.

Note: This report describes the analysis workflow. Actual product names and sales values are not included because the notebook results have not been executed and verified for this report.